

Namespace Manager - Data Structure Reference

Overview

This module implements the **Namespace & Result Manager** role in the system architecture. It manages MATLAB workspace snapshots, runtime-level variable tracking, execution logs, and provides structured outputs for debugging and workflow tracing.

1. Module-Level Attributes

`current_snapshot: dict[str, dict]`

Latest workspace snapshot (not currently updated in code).

Format:

```
{
  "<var_name>": {
    "class": "<MATLAB class name>",
    "size": "<rows>x<cols>"
  }
}
```

Example:

```
{
  "topo": { "class": "double", "size": "256x256" },
  "params": { "class": "struct", "size": "1x1" },
  "filename": { "class": "char", "size": "1x12" }
}
```

`runtime_snapshots: dict[str, dict]`

Stores **start** and **end** workspace snapshots for each runtime.

Format:

```
{
  "<runtime_EDID>": {
    "start": { "<var>": { "class": "...", "size": "..." } },
    "end": { "<var>": { "class": "...", "size": "..." } } | null
  }
}
```

Example:

```
{
  "runtime_OP_1": {
    "start": {
      "raw_path": {"class": "char", "size": "1x28"}
    },
    "end": null
  }
}
```

runtime_outputs: dict[str, dict]

Stores the **diff** (new/modified variables) calculated at the end of a runtime.

Format:

```
{
  "<runtime_EDID>": {
    "<var_name>": {
      "class": "<MATLAB class name>",
      "size": "<rows>x<cols>",
      "status": "added" | "modded"
    }
  }
}
```

Example:

```
{
  "runtime_OP_1": {
    "topo": {"class": "double", "size": "256x256", "status": "added"},
    "fft": {"class": "double", "size": "256x256", "status": "added"}
  }
}
```

execution_log: list[dict]

Global script call log (not tied to a specific runtime).

Format:

```
[
  {"script": "<script_name>", "params": { /* parameters */ }},
  ...
]
```

Example:

```
[
  {"script": "opensxm_auto", "params": {"path": "Data", "filename":
"07_15_25_002.sxm", "pnum": "1"}},
  {"script": "polybackRow", "params": {"figure_name": "fig_07_15_25_002",
"order": 3}}
]
```

`runtime_logs: dict[str, list[dict]]`

Runtime-specific script call logs.

Format:

```
{
  "<runtime_EDID>": [
    {"script": "<script_name>", "params": { /* parameters */ }},
    ...
  ]
}
```

Example:

```
{
  "runtime_OP_1": [
    {"script": "opensxm_auto", "params": {"path": "Data", "filename":
"07_15_25_002.sxm", "pnum": "1"}},
    {"script": "fftamp", "params": {"input_var": "topo"}}
  ]
}
```

2. Function Outputs

`snapshot_workspace() -> dict[str, dict]`

Returns the current workspace snapshot.
See `current_snapshot` format for structure.

`start_runtime(runtime_EDID: str) -> None`

Stores the start snapshot under:

```
runtime_snapshots[runtime_EDID] = {  
    "start": <snapshot>,  
    "end": None  
}
```

`log_script_call(script_name: str, params: dict, runtime_EDID: str|None) -> None`

Appends a log entry to:

- `execution_log` (global)
- `runtime_logs[runtime_EDID]` (if provided)

Log entry format:

```
{"script": "<script_name>", "params": { /* parameters */ }}
```

`end_runtime(runtime_EDID: str) -> dict`

Calculates the diff between start and end snapshots.

Updates:

```
runtime_snapshots[runtime_EDID]["end"] = end_snapshot  
runtime_outputs[runtime_EDID] = diff
```

Return format:

```
{  
    "diff": { "<var_name>": {"class": "...", "size": "...", "status": "..."} },  
    "log": [ {"script": "...", "params": {...}}, ... ]  
}
```

`get_runtime_outputs(runtime_EDID: str) -> dict`

Returns:

```
{
  "diff": { /* same as runtime_outputs[runtime_EDID] */ },
  "log": [ /* same as runtime_logs[runtime_EDID] */ ]
}
```

`get_runtime_status(runtime_EDID: str) -> str`

Possible values: "pending" | "running" | "success" | "failed" | "unknown"

3. Full Runtime Example

```
{
  "runtime_snapshots": {
    "runtime_OP_1": {
      "start": {
        "raw_path": {"class": "char", "size": "1x28"}
      },
      "end": {
        "raw_path": {"class": "char", "size": "1x28"},
        "topo": {"class": "double", "size": "256x256"},
        "fft": {"class": "double", "size": "256x256"}
      }
    }
  },
  "runtime_logs": {
    "runtime_OP_1": [
      {"script": "opensxm_auto", "params": {"path": "Data", "filename":
"07_15_25_002.sxm", "pnum": "1"}},
      {"script": "fftamp", "params": {"input_var": "topo"}}
    ]
  },
  "runtime_outputs": {
    "runtime_OP_3": {
      "topo": {"class": "double", "size": "256x256", "status": "added"},
      "fft": {"class": "double", "size": "256x256", "status": "added"}
    }
  },
  "execution_log": [
    {"script": "opensxm_auto", "params": {"path": "Data", "filename":
"07_15_25_002.sxm", "pnum": "1"}},
    {"script": "fftamp", "params": {"input_var": "topo"}}
  ]
}
```

